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Safety data sheet

BASF Safety data sheet according to UN GHS 4th rev.
Date / Revised: 02.02.2023 Version: 4.0
Product: **Basolon GL 40 exp**
(ID no. 30735178/SDS_GEN_00/EN)
Date of print 05.07.2025

1. Identification

Product identifier

Basolon GL 40 exp

Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Performance Chemicals for Oilfield Applications

Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Global Oilfield Chemicals

Telephone: +49 621 60-0
E-mail address: RegXcellenceOilfieldChemicals@basf.com

Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

Acute Tox. 5 (oral)
Acute Tox. 4 (Inhalation - mist)
Skin Corr./Irrit. 2
Eye Dam./Irrit. 2B

Skin Sens. 1
STOT SE 3 (irritating to respiratory system)
Aquatic Acute 3

For the classifications not written out in full in this section the full text can be found in section 16.

Label elements

Globally Harmonized System (GHS)

Pictogram:

Signal Word:
Warning

Hazard Statement:

H320	Causes eye irritation.
H315	Causes skin irritation.
H332	Harmful if inhaled.
H303	May be harmful if swallowed.
H317	May cause an allergic skin reaction.
H335	May cause respiratory irritation.
H402	Harmful to aquatic life.

Precautionary Statements (Prevention):

P280	Wear protective gloves.
P271	Use only outdoors or in a well-ventilated area.
P260	Do not breathe mist or vapour.
P273	Avoid release to the environment.
P272	Contaminated work clothing should not be allowed out of the workplace.
P264	Wash contaminated body parts thoroughly after handling.

Precautionary Statements (Response):

P312	Call a POISON CENTER or physician if you feel unwell.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P302 + P352	IF ON SKIN: Wash with plenty of soap and water.
P333 + P313	If skin irritation or rash occurs: Get medical attention.
P332 + P313	If skin irritation occurs: Get medical attention.
P362 + P364	Take off contaminated clothing and wash it before reuse.
P337 + P313	If eye irritation persists: Get medical attention.

Precautionary Statements (Storage):

P403 + P233	Store in a well-ventilated place. Keep container tightly closed.
P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

According to UN GHS criteria

Hazard determining component(s) for labelling: Glyoxal

Other hazards

According to UN GHS criteria

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

3. Composition/Information on Ingredients

Substances

Not applicable

Mixtures

Chemical nature

mixture

Hazardous ingredients (GHS)
According to UN GHS criteria

Glyoxal

Content (W/W): >= 39,5 % - <= 40,5 % CAS Number: 107-22-2 EC-Number: 203-474-9	Acute Tox. 5 (oral) Acute Tox. 4 (Inhalation - mist) Skin Corr./Irrit. 2 Eye Dam./Irrit. 2B Skin Sens. 1 STOT SE 3 (irr. to respiratory syst.) Aquatic Acute 3 H320, H315, H332, H303, H317, H335, H402

Ethane-1,2-diol

Content (W/W): >= 0 % - <= 2,5 % CAS Number: 107-21-1 EC-Number: 203-473-3 INDEX-Number: 603-027-00-1	Acute Tox. 4 (oral) STOT RE (Kidney) 2 H302, H373

Glycollic acid

Content (W/W): >= 0,05 % - <= 0,35 % CAS Number: 79-14-1 EC-Number: 201-180-5	Acute Tox. 4 (oral) Acute Tox. 4 (Inhalation - mist) Skin Corr./Irrit. 1B Eye Dam./Irrit. 1 Aquatic Acute 3 H314, H302 + H332, H402 EUH071

Formaldehyde

Content (W/W): >= 0 % - <= 0,01 % CAS Number: 50-00-0 EC-Number: 200-001-8 INDEX-Number: 605-001-00-5	Flam. Liq. 4 Acute Tox. 2 (Inhalation - vapour) Acute Tox. 3 (oral) Acute Tox. 3 (dermal) Skin Corr./Irrit. 1B Eye Dam./Irrit. 1 Skin Sens. 1A Carc. 1B Aquatic Acute 2 H227, H330, H317, H350, H314, H301 + H311, H401 <u>Specific concentration limit:</u> Skin Corr./Irrit. 1B: >= 25 % Skin Sens. 1: >= 0,2 % Skin Corr./Irrit. 2: 5 - < 25 % STOT SE 3, irr. to respiratory syst.: >= 5 % Eye Dam./Irrit. 2: 5 - < 25 %

For the classifications not written out in full in this section the full text can be found in section 16.

4. First-Aid Measures

Description of first aid measures

Remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
water spray, foam, dry powder, carbon dioxide

Special hazards arising from the substance or mixture

nitrogen oxides, carbon oxides
The substances/groups of substances mentioned can be released in case of fire.

Advice for fire-fighters

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems.

6. Accidental Release Measures

Personal precautions, protective equipment and emergency procedures

Avoid contact with the skin, eyes and clothing.

Environmental precautions

Do not discharge into drains/surface waters/groundwater.

Methods and material for containment and cleaning up

For large amounts: Pump off product.
For residues: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr). Dispose of absorbed material in accordance with regulations.

7. Handling and Storage

Precautions for safe handling

Ensure thorough ventilation of stores and work areas.

Conditions for safe storage, including any incompatibilities

Suitable materials for containers: Stainless steel 1.4401, Stainless steel 1.4301 (V2), High density polyethylene (HDPE), glass, Low density polyethylene (LDPE)
Unsuitable materials for containers: Paper/Fibreboard
Further information on storage conditions: Protect from air.

Storage stability:
Storage temperature: < 50 °C
Storage duration: 6 Months
May yellow after lengthy storage.
During the storage a product characteristic clouding or crystallin precipitation of the trimeric glyoxal hydrate may occur.
The process is reversible if warmed up mild (max. 40 °C).
From the data on storage duration in this safety data sheet no agreed statement regarding the warrantee of application properties can be deduced.

Protect from temperatures above: 50 °C
The packed product must be protected against exceeding the indicated temperature.

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

50-00-0: Formaldehyde
107-21-1: Ethane-1,2-diol
107-22-2: Glyoxal

Exposure controls

Personal protective equipment

Respiratory protection:
Wear respiratory protection if ventilation is inadequate. Gas filter for gases/vapours of organic compounds (boiling point <65 °C, f.e. EN 14387 Type AX)

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc. Manufacturer's directions for use should be observed because of great diversity of types.

Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
Avoid contact with the skin, eyes and clothing. When using, do not eat, drink or smoke. Avoid contact with skin and eyes. Remove soiled or soaked clothing immediately. Wash off any contamination that gets onto the skin with plenty of water and soap, skin care.

9. Physical and Chemical Properties

Information on basic physical and chemical properties

Form:	liquid	
Colour:	colourless to yellow	
Odour:	aldehyde-like	
Odour threshold:		
	Not determined due to potential health hazard by inhalation.	
pH value:	2 - 3 (40 %(m), 20 °C)	
melting range:	-50 - -15 °C	(OECD Guideline 102)
Boiling point:	103,6 °C (1.013,25 hPa)	(Directive 92/69/EEC, A.2)
Flash point:	> 93 °C	(ISO 13736)
Evaporation rate:		
	Value can be approximated from Henry's Law Constant or vapor pressure.	
Flammability:	not flammable	(derived from flash point)
Lower explosion limit:		
	For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point.	
Upper explosion limit:		
	For liquids not relevant for classification and labelling.	
Ignition temperature:	285 °C	(DIN 51794)
Vapour pressure:	20,2 hPa (20 °C) Information applies to the solvent.	(Directive 92/69/EEC, A.4)
	106,7 hPa (50 °C) Information applies to the solvent.	(Directive 92/69/EEC, A.4)
Density:	approx. 1,27 g/cm3 (40 %(m), 20 °C) 1,2514 g/cm3 (50 °C)	(Directive 92/69/EEC, A.3)
Relative density:	1,27 (20 °C)	(Directive 92/69/EEC, A.3)
Relative vapour density (air):		
	Water content greater than 10%.	
Solubility in water:	miscible (20,2 °C)	(internal method)
Partitioning coefficient n-octanol/water (log Kow):	-1,15 (23 °C; pH value: 7)	(Directive 92/69/EEC, A.8)
	-1 (23 °C; pH value: 5)	(Directive 84/449/EEC, A.8)
	-1,62 (23 °C; pH value: 9)	(Directive 84/449/EEC, A.8)

Self ignition:	Based on its structural properties the product is not classified as self-igniting.	Test type: Spontaneous self-ignition at room-temperature.
Thermal decomposition:	No decomposition if correctly stored and handled.	
Viscosity, dynamic:	8,37 mPa.s (20 °C)	(OECD 114)
	4,25 mPa.s (40 °C)	(OECD 114)
Viscosity, kinematic:	6,6 mm2/s (20 °C)	(OECD 114)
	3,38 mm2/s (40 °C)	(OECD 114)
Explosion hazard:	Based on the chemical structure there is no indication of explosive properties.	
Fire promoting properties:	Based on its structural properties the product is not classified as oxidizing.	

Other information

Self heating ability:	It is not a substance capable of spontaneous heating.	
pKA:	The substance does not dissociate.	
Adsorption/water - soil:	KOC: 2,1; log KOC: 0,32	(OECD Guideline 121)
Surface tension:	Based on chemical structure, surface activity is not to be expected.	
Grain size distribution:	The substance / product is marketed or used in a non solid or granular form.	
Molar mass:	58,04 g/mol	

10. Stability and Reactivity

Reactivity
No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals:	No corrosive effect on metal.	
Formation of flammable gases:	Remarks:	Forms no flammable gases in the presence of water.

Chemical stability
The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions
Reacts with strong alkalies. Exothermic reaction.

Conditions to avoid
Temperature: > 50 °C

See SDS section 7 - Handling and storage.

Incompatible materials

Substances to avoid:
strong alkalies

Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Of low toxicity after single ingestion. Virtually nontoxic after a single skin contact. Of moderate toxicity after short-term inhalation.

Experimental/calculated data:
LD50 rat (oral): 3.300 mg/kg (OECD Guideline 401)

LC50 rat (by inhalation): 2,44 mg/l 4 h (OECD Guideline 403)
An aerosol was tested.

LD50 rat (dermal): > 2.000 mg/kg (OECD Guideline 402)
Limit concentration test only (LIMIT test).

Irritation

Assessment of irritating effects:
Eye contact causes irritation. Skin contact causes irritation.

Experimental/calculated data:
Skin corrosion/irritation rabbit: Irritant. (OECD Guideline 404)

Serious eye damage/irritation rabbit: Irritant. (OECD Guideline 405)

Respiratory/Skin sensitization

Assessment of sensitization:
Caused skin sensitization in animal studies. Caused sensitization in humans.

Experimental/calculated data:
Guinea pig maximization test guinea pig: skin sensitizing (OECD Guideline 406)

Human Maximization Test human: skin sensitizing (Patch test)
Literature data.

Germ cell mutagenicity

Assessment of mutagenicity:
The substance was mutagenic in various test systems with microorganisms and cell cultures; however, these results could not be confirmed in tests with mammals.

Carcinogenicity

Assessment of carcinogenicity:
The whole of the information assessable provides no indication of a carcinogenic effect.

Reproductive toxicity

Assessment of reproduction toxicity:
The results of animal studies gave no indication of a fertility impairing effect.

Developmental toxicity

Assessment of teratogenicity:
No indications of a developmental toxic / teratogenic effect were seen in animal studies.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Causes temporary irritation of the respiratory tract.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
After repeated exposure the prominent effect is local irritation.

Aspiration hazard

No aspiration hazard expected.

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:
Acutely harmful for aquatic organisms. Based on long-term (chronic) toxicity study data, the product is very likely not harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Toxicity to fish:
LC50 (96 h) > 186 - < 272 mg/l, Leuciscus idus (DIN 38412 Part 15, static)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

Aquatic invertebrates:
EC50 (48 h) 161 mg/l, Daphnia magna (Directive 79/831/EEC, static)
Nominal concentration. The ecological data given are those of the active ingredient.

LC50 (96 h) 76,0 mg/l, Americamysis bahia (other, semistatic)

The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

Aquatic plants:
EC50 (72 h) > 40 mg/l (growth rate), Scenedesmus subspicatus (OECD Guideline 201, static)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

EC10 (72 h) > 10 mg/l (growth rate), Scenedesmus subspicatus (OECD Guideline 201, static)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

EC50 (72 h) 347,1 mg/l (growth rate), Skeletonema costatum (OECD Guideline 201)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

No observed effect concentration (72 h) 118,4 mg/l (growth rate), Skeletonema costatum (OECD Guideline 201)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

Microorganisms/Effect on activated sludge:
EC10 (16 h) 22,8 mg/l, Pseudomonas putida (DIN 38412 Part 8, static)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

EC50 (0,5 h) > 400 mg/l, activated sludge (OECD Guideline 209, static)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

Chronic toxicity to fish:
No observed effect concentration (34 d) 112 mg/l, Pimephales promelas (OPP 72-4 (EPA-Guideline), Flow through.)
The ecological data given are those of the active ingredient. The statement of the toxic effect relates to the analytically determined concentration.

Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d) 3,19 mg/l, Daphnia magna (OECD Guideline 211, semistatic)
The statement of the toxic effect relates to the analytically determined concentration. The ecological data given are those of the active ingredient.

Assessment of terrestrial toxicity:
Toxic effects have been observed in studies with soil living organisms. Toxic effects have been observed in studies with terrestric plants.

Soil living organisms:
LC50 (14 d) > 135 mg/kg, Eisenia foetida (OECD Guideline 207, artificial soil)
The ecological data given are those of the active ingredient. The details of the toxic effect relate to the nominal concentration.

EC50 (28 d) > 1.054 mg/kg, soil dwelling microorganisms (OECD 217, natural soil)
The details of the toxic effect relate to the nominal concentration. The data have been calculated from values for a preparation with a lower substance concentration. The ecological data given are those of the active ingredient.

EC50 (28 d) > 1.054 mg/kg, soil dwelling microorganisms (OECD 216, natural soil)
The details of the toxic effect relate to the nominal concentration. The data have been calculated from values for a preparation with a lower substance concentration. The ecological data given are those of the active ingredient.

Terrestrial plants:
No observed effect concentration (21 d) 314 mg/kg, Brassica napus (OECD Guideline 208)
The ecological data given are those of the active ingredient. The data have been deduced from values for a preparation or mixture with a lower substance concentration.

Other terrestrial non-mammals:
No data available.

Persistence and degradability

Assessment biodegradation and elimination (H2O):
Readily biodegradable (according to OECD criteria).

Elimination information:
90 - 100 % DOC reduction (19 d) (OECD 301 A (new version)) (aerobic, activated sludge, domestic, non-adapted) Readily biodegradable (according to OECD criteria).

Assessment of stability in water:
According to structural properties, hydrolysis is not expected/probable.
Information on Stability in Water (Hydrolysis):
The product has not been tested. The statement has been derived from the structure of the product.

Bioaccumulative potential

Assessment bioaccumulation potential:
Significant accumulation in organisms is not to be expected.

Bioaccumulation potential:
Bioconcentration factor: 3,2, Fish (calculated)

Mobility in soil

Assessment transport between environmental compartments:
Volatility: The substance will not evaporate into the atmosphere from the water surface.
Adsorption in soil: Adsorption to solid soil phase is not expected.

Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not fulfill the criteria for PBT (Persistent/bioaccumulative/toxic) and vPvB (very persistent/very bioaccumulative). Self classification

Other adverse effects

The substance is not listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

Additional information

Sum parameter

Chemical oxygen demand (COD): 350 mg/g

Biochemical oxygen demand (BOD) Incubation period 5 d: 175 mg/g

13. Disposal Considerations

Waste treatment methods

Incinerate in suitable incineration plant, observing local authority regulations.

A waste code in accordance with the European waste catalog (EWC) cannot be specified, due to dependence on the usage.
The waste code in accordance with the European waste catalog (EWC) must be specified in cooperation with disposal agency/manufacturer/authorities.

Contaminated packaging:
Uncontaminated packaging can be recycled.

14. Transport Information

Land transport

ADR

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

RID

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Inland waterway transport
ADN

Not classified as a dangerous good under transport regulations

UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user:	None known

Transport in inland waterway vessel

UN number or ID number:	ID9006
UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (GLYOXAL)
Transport hazard class(es):	9, N3
Packing group:	Not applicable
Environmental hazards:	yes
Type of inland waterway vessel:	N
Cargo tank design:	4
Cargo tank type:	2

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Maritime transport in bulk according to IMO instruments

Regulation:	IBC-Code

Product name:	Glyoxal solution (40% or less)
Pollution category:	Y
Ship Type:	3

15. Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

Annex XVII of Regulation (EC) No 1907/2006
Number on List: 29

16. Other Information

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

Acute Tox.	Acute toxicity
Skin Corr./Irrit.	Skin corrosion/irritation
Eye Dam./Irrit.	Serious eye damage/eye irritation
Skin Sens.	Skin sensitization
STOT SE	Specific target organ toxicity — single exposure
Aquatic Acute	Hazardous to the aquatic environment - acute
STOT RE	Specific target organ toxicity — repeated exposure
Flam. Liq.	Flammable liquids
Carc.	Carcinogenicity
H320	Causes eye irritation.
H315	Causes skin irritation.
H332	Harmful if inhaled.
H303	May be harmful if swallowed.
H317	May cause an allergic skin reaction.
H335	May cause respiratory irritation.
H402	Harmful to aquatic life.
H302	Harmful if swallowed.
H373	May cause damage to organs (Kidney) through prolonged or repeated exposure.
H314	Causes severe skin burns and eye damage.
H302 + H332	Harmful if swallowed or if inhaled
H227	Combustible liquid.
H330	Fatal if inhaled.
H350	May cause cancer.
H301 + H311	Toxic if swallowed or in contact with skin
H401	Toxic to aquatic life.
EUH071	Corrosive to the respiratory tract.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification

agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.